PROPHET: Phylogenetically Robust Binder Design Against Heterogeneous Evolutionary Trajectories

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Abstract

Antiviral therapeutics that bind a pathogen's current surface proteins inevitably lose efficacy as the virus accumulates escape mutations, yet existing binder design methods optimize binding against a single observed sequence and have no mechanism to reason about evolutionary futures. We introduce PROPHET (**P**hylogenetically **R**obust **O**ptimization for **P**eptide design against **H**eterogeneous **E**volutionary **T**rajectories), a framework that performs robust antiviral binder design in observed viral phylogenies. Given a collection of FastTree phylogenies over a target viral protein, we estimate a site-specific phylogenetic energy function combining per-position mutation rates, empirical substitution matrices, and direct coupling analysis of coevolving residues, defining a biologically meaningful distribution over evolutionarily plausible future variants without requiring a learned generative model of tree topology. Escape-robust peptides are then obtained by guiding multi-objective discrete flow matching against the worst-case binding loss over the implied variant distribution, instantiated as a probability-weighted expectation over Gibbs-sampled sequences drawn from the phylogenetic energy landscape. The resulting binders are optimized simultaneously for affinity to the current variant and binding retention across the full distribution of plausible evolutionary futures implied by observed phylogenetic dynamics. On HIV-1 protease, where resistance mutation landscapes are well characterized, PROPHET produces peptides with significantly higher mean and minimum binding scores across held-out escape variants than baselines optimizing against the wildtype alone or against uniformly weighted observed leaf sequences, demonstrating that phylogenetically informed distributional robustness is both necessary and sufficient to close the gap between single-target binder design and evolution-proof antiviral therapeutics.

1 Introduction

Viral proteins are moving targets. HIV-1 protease accumulates resistance mutations under antiretroviral pressure within weeks of treatment initiation (Apetroaei et al., 2024), influenza hemagglutinin drifts antigenically across seasons (Hensley et al., 2009), and SARS-CoV-2 spike has repeatedly acquired variants that erode the efficacy of antibody-based therapeutics (Sharun et al., 2021). Peptide therapeutics are among the most rapidly deployable countermeasures against emerging and evolving viral threats: they can be computationally designed in hours, synthesized in days, and manufactured at scale without the cell-line development timelines required by monoclonal antibodies (Muttenthaler et al., 2021; Wang et al., 2022; Hong et al., 2026). Yet this speed advantage is squandered when the designed peptide is optimized against a single snapshot of a target that will have mutated by the time treatment reaches patients. The evolutionary trajectory that led to escape was, in many cases, predictable from the virus's own phylogenetic history, and the central open problem is designing peptides whose binding remains effective across the distribution of variants a virus is likely to explore. Current computational design methods view the target as a fixed sequence or structure, optimize affinity against it, and provide no mechanism to account for future evolutionary variation.

Generative approaches to peptide and protein binder design have advanced rapidly. Structure-based methods (Watson et al., 2023; Pacesa et al., 2025; Stark et al., 2025) produce high-affinity binders against static targets but inherit an equilibrium assumption that provides no handle on evolutionary robustness. Discrete diffusion and discrete flow models (Austin et al., 2021; Sahoo et al., 2024; Shi et al., 2024; Lou et al., 2024; Gat et al., 2024) have emerged as a powerful alternative for biological sequence design, and reward-guided fine-tuning and multi-objective guidance frameworks (Tang

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et al., 2025a;b; Wang et al., 2025; Cao et al., 2025; Chen et al., 2025b;c) allow these models to optimize sequences, including peptides, against arbitrary objectives, including affinity, developability, and specificity (Zhang et al., 2026; Chen et al., 2025a; Vincoff et al., 2025). Still, every such framework defines the design objective against a single fixed target sequence. Extending guidance to a *distribution* over targets, shaped by genuine evolutionary dynamics, requires both a principled model of that distribution and a tractable way to optimize against it. Neither exists for antiviral design.

We introduce PROPHET (**P**hylogenetically **R**obust **O**ptimization for **P**eptide design against **H**eterogeneous **E**volutionary **T**rajectories), a framework that builds evolutionary robustness directly into the discrete flow matching design objective by basing variant uncertainty in observed viral phylogenies. Given hundreds of FastTree phylogenies over a target viral protein (Price et al., 2009), PROPHET estimates a site-specific phylogenetic energy function from per-position mutation rates, empirical substitution matrices, and direct coupling analysis of coevolving residues. This energy function defines a biologically constrained distribution over plausible future variants without requiring a learned generative model of tree topology. We then guide multi-objective discrete flow matching (Chen et al., 2025c) against the worst-case binding loss over Gibbs-sampled sequences from this distribution, producing peptides that simultaneously optimize affinity to the current variant, drug-like properties, and binding retention across the full phylogenetically implied variant landscape. Our contributions are as follows:

- **Phylogenetic energy function for variant uncertainty.** We introduce a site-specific evolutionary energy model estimated directly from FastTree phylogenies (Price et al., 2009), combining per-position mutation rates, substitution preferences, and direct residue couplings into a principled distribution over future variants. The sublevel sets of this energy define a biologically meaningful uncertainty set that respects the actual geometry of viral sequence evolution rather than imposing an arbitrary ball in sequence space.
- **Distributional robustness via phylogenetic Gibbs sampling.** We formulate escape-robust peptide design as optimization against the worst-case binding loss over the phylogenetic variant distribution, instantiated by Gibbs sampling sequences from the energy landscape. This reduces evolutionary robustness to a tractable multi-objective guidance signal compatible with any pre-trained discrete flow model, requiring no architectural change and no new experimental data.
- **Unified multi-objective design against evolutionary futures.** We integrate phylogenetic robustness as a fourth guidance objective within the MOG-DFM (Chen et al., 2025c) framework alongside wildtype binding affinity, motif specificity, and drug-like properties, yielding Pareto-efficient peptides that trade off single-target potency against broad-spectrum evolutionary robustness in a principled and controllable way.
- **Empirical validation on HIV-1 protease.** We demonstrate on HIV-1 protease, where resistance mutation landscapes are well characterized (Humphris-Narayanan et al., 2012), that PROPHET peptides achieve significantly higher mean and minimum binding scores across held-out escape variants than baselines optimizing against the wildtype alone or against uniformly weighted observed leaf sequences, while incurring only a modest reduction in wildtype affinity.

2 Preliminaries

2.1 Multi-Objective Guided Discrete Flow Matching

For discrete data $x \in \mathcal{S} = \mathcal{T}^d$ where $\mathcal{T} = [K]$ is the amino acid vocabulary, discrete flow matching (Gat et al., 2024) models a continuous-time Markov chain $\{X_t\}_{t\in[0,1]}$ with transition rates $u_t(y,x)$ evolving marginal probabilities via the Kolmogorov forward equation $\frac{d}{dt}p_t(y) = \sum_x u_t(y,x)\, p_t(x)$. The velocity field factorizes over positions as $u_t(y,x) = \sum_i \delta(y^{\bar{i}}, x^{\bar{i}})\, u_t^i(y^i, x)$ and is trained via conditional flow matching:

$$\mathcal{L}_{\mathrm{CDFM}}(\theta) = \mathbb{E}_{t,\,Z,\,X_t\sim p_{t|Z}} \sum_i D^i_{X_t}\big(u_t^i(\cdot, X_t|Z),\; u^i_{\theta,t}(\cdot, X_t)\big), \quad (1)$$

where $D^i_{X_t}$ is a Bregman divergence.

MOG-DFM (Chen et al., 2025c) steers pre-trained discrete flow models toward Pareto-efficient solutions across multiple objectives $\{s_n\}_{n=1}^N$. At each sampling step, guided transition rates combine rank-based improvement with directional alignment toward a weight vector ω. For $y^i \neq x^i$:

$$\tilde{u}_t^i(y^i, x \mid \omega) = \beta\, u_t^i(y^i, x)\, \exp\big(\Delta S(y^i, x, \omega)\big), \quad (2)$$

with the diagonal entry set to $\tilde{u}_t^i(x^i, x \mid \omega) = -\sum_{y^i\neq x^i} \tilde{u}_t^i(y^i, x \mid \omega)$ to conserve probability. Here ΔS combines normalized rank scores with a directional term $\Delta s(y^i, x)\cdot\omega$, and adaptive hypercone filtering restricts transitions to lie within an angle Φ around ω. MOG-DFM is agnostic to the objectives $\{s_n\}$, which we exploit by introducing a phylogenetically derived robustness objective.

2.2 Phylogenetic Trees and Substitution Models

A rooted phylogenetic tree $T = (V, E, \{b_e\})$ consists of a topology (V, E) and branch lengths $\{b_e\}$ representing evolutionary distance. Given a multiple sequence alignment (MSA) of observed viral protein sequences, approximate maximum-likelihood phylogenies are inferred using FastTree (Price et al., 2009). A collection $\{T_j\}_{j=1}^{J}$ of phylogenies over the same protein family captures uncertainty in both topology and rate estimates.

From a tree ensemble, we estimate site-specific parameters at each alignment position i: a *mutation rate* λ_i (ratio of observed substitutions to total branch length) and a *substitution matrix* $Q_i \in \mathbb{R}^{20\times 20}$ recording empirical amino acid transition frequencies, with rows normalized to yield conditional distributions. Conserved positions exhibit low λ_i and concentrated Q_i; variable surface-exposed positions exhibit high λ_i and diffuse Q_i.

2.3 Direct Coupling Analysis

Site-specific parameters capture marginal evolutionary statistics but miss correlated mutations between positions. Direct coupling analysis (DCA) (Morcos et al., 2011; Ekeberg et al., 2013) infers pairwise couplings from an MSA by fitting a Potts model:

$$p_{\text{Potts}}(x) = \frac{1}{Z}\,\exp\Bigg(\sum_i h_i(x_i) + \sum_{i<j} J_{ij}(x_i, x_j)\Bigg), \quad (3)$$

where $h_i(x_i)$ are site-specific fields and $J_{ij}(x_i, x_j)$ are pairwise couplings estimated by pseudolikelihood maximization (Ekeberg et al., 2013). Strong couplings identify co-mutating positions linked by structural contacts or functional dependencies. In viral evolution, DCA couplings capture epistatic constraints: certain resistance mutations co-occur to compensate for fitness costs, while others are mutually exclusive.

3 Problem Formulation

Existing multi-objective peptide design (§2.1) optimizes binding against a *fixed* target sequence and has no mechanism for evolutionary robustness. We extend the framework by (i) defining a phylogenetically informed distribution over future variants, and (ii) introducing an evolutionary robustness objective that guides discrete flow matching against worst-case binding loss under this distribution.

3.1 Phylogenetic Energy Function

We unify site-specific evolutionary parameters (§2.2) and pairwise couplings (§2.3) into a single energy over viral protein sequences. Given wildtype $x_{\text{WT}} \in \mathcal{T}^N$ and phylogenetic parameters $(\{\lambda_i\}, \{h_i\}, \{J_{ij}\})$, the *phylogenetic energy* of a variant x is:

$$E_{\text{evo}}(x) = \sum_{i=1}^{N} \lambda_i\, h_i(x_i) + \sum_{i<j} J_{ij}(x_i, x_j), \quad (4)$$

where $h_i(x_i)$ encodes site preferences weighted by mutation rate λ_i, and J_{ij} encodes epistatic constraints. The λ_i weighting ensures that conserved positions contribute little to energy differences between sequences, while rapidly evolving positions dominate. This defines a Boltzmann distribution:

$$p_{\text{evo}}(x) = \frac{1}{Z_{\text{evo}}}\exp\big(-E_{\text{evo}}(x)/T_{\text{evo}}\big), \quad (5)$$

with temperature $T_{\text{evo}} > 0$ controlling breadth. We calibrate T_{evo} so that sampled edit distances match those observed across leaf sequences in held-out phylogenies (§4).

3.2 Gibbs Sampling of Future Variants

Since Z_{evo} is intractable, we sample from p_{evo} via Gibbs sampling. Starting from x_{WT}, we resample positions from their conditionals:

$$p_{\text{evo}}(x_i|x_{\setminus i}) \propto \exp\Bigg(-\frac{\lambda_i h_i(x_i) + \sum_{j\neq i} J_{ij}(x_i, x_j)}{T_{\text{evo}}}\Bigg), \quad (6)$$

a categorical distribution over 20 amino acids, trivially normalized at each step. After burn-in, we collect M samples $\{x^{(m)}\}_{m=1}^{M}$.

To ensure biological plausibility, we filter samples using ESM-2 (Lin et al., 2023) pseudo-log-likelihood:

$$\text{pLL}(x) = \sum_i \log p_{\text{ESM}}(x_i|x_{\setminus i}) \geq \text{pLL}(x_{\text{WT}}) - \delta, \quad (7)$$

rejecting variants that are phylogenetically plausible but structurally unviable.

3.3 Evolutionary Robustness Objective

Given sampled variants $\mathcal{V} = \{x^{(m)}\}_{m=1}^{M}$ from p_{evo}, we define robustness via conditional value at risk (CVaR):

$$s_{\text{robust}}(y;\,\eta) = \frac{1}{\lfloor \eta M \rfloor}\sum_{m=1}^{\lfloor \eta M \rfloor} \text{Aff}(y,\, x^{(\sigma(m))}), \quad (8)$$

where σ sorts variants by ascending binding score and $\eta \in (0, 1]$ controls worst-case stringency (η=1 recovers the mean; $\eta \to 0$ approaches the minimum).

3.4 Escape-Robust Peptide Design

We integrate robustness into MOG-DFM alongside wildtype binding:

$$s_1(y) = \text{Aff}(y,\, x_{\text{WT}}), \quad (9)$$
$$s_2(y) = s_{\text{robust}}(y;\,\eta), \quad (10)$$

where s_1 scores affinity to the current variant and s_2 is phylogenetic robustness (Eq. 8). Guided transition rates

(Eq. 2) use both objectives with weight vector $\omega \in \Delta^1$, enabling controlled navigation of the trade-off between current-variant potency and evolutionary robustness.

The key distinction from prior work is that s_2 is an expectation over a phylogenetically grounded variant distribution, coupling guidance to observed evolutionary dynamics rather than treating robustness as an ad hoc regularizer or a static set of known variants.

3.5 Design Task

Design Task. Given a viral target x_{WT} and phylogenies $\{T_j\}_{j=1}^{J}$, generate a peptide $y \in \mathcal{A}^L$ that achieves high binding to x_{WT} while maintaining binding across evolutionarily plausible future variants implied by p_{evo}, as measured by $s_{\mathrm{robust}}(y;\eta)$.

This extends single-target design to distributional robustness over evolutionary futures, requiring no learned tree generator, no adversarial training, and no data beyond input phylogenies and a pre-trained binding predictor.

4 Results

We evaluate PROPHET along five axes: (1) quality of phylogenetically sampled variants, (2) escape-robust peptide design against held-out escape variants, (3) analysis of the robustness–affinity Pareto front, (4) ablations isolating each component, and (5) sensitivity to key hyperparameters.

All experiments target HIV-1 protease (UniProt P04585) as a canonical model of evolutionary escape, given its well-characterized resistance mutation landscape and clinical relevance to antiretroviral therapy failure (Humphris-Narayanan et al., 2012; Kneller et al., 2020). We use PeptiVerse (Zhang et al., 2026) as the binding affinity predictor throughout, with experimentally-correlated scores normalized between $[0, 1]$ where higher indicates stronger predicted binding.

Phylogenetic data. We construct phylogenies from HIV-1 protease sequences obtained from the Los Alamos HIV Sequence Database. After filtering for full-length, non-redundant protease sequences, we build $J = 200$ FastTree (**?**) phylogenies from bootstrap-resampled MSAs. Site-specific mutation rates $\{\lambda_i\}$, substitution matrices $\{Q_i\}$, and DCA couplings $\{h_i, J_{ij}\}$ are estimated from the full MSA as described in §2.2–§2.3. We hold out 20% of phylogenies for calibrating T_{evo} and evaluating variant quality.

Baselines. We compare PROPHET against six baselines:

1. **RFdiffusion** (Watson et al., 2023): structure-based binder design against the wildtype crystal structure, with no sequence-level evolutionary information.
2. **MOG-DFM (WT only):** optimizes peptides solely against the wildtype with no robustness objective.
3. **MOG-DFM (uniform leaves):** two-objective formulation with s_2 defined as the mean binding over uniformly weighted observed leaf sequences.
4. **MOG-DFM (random var.):** s_2 defined over randomly mutated variants, matched in edit distance to PROPHET samples.
5. **MOG-DFM (ESM-only var.):** s_2 using variants from iterative ESM-2 masked prediction, matched in edit distance.
6. **PepTune** (Tang et al., 2025a): multi-objective guided discrete diffusion without evolutionary robustness.

Table 1. Variant quality. Res. enr. is resistance site enrichment. pLL is ESM-2 pseudo-log-likelihood. DCA E is mean Potts energy.

Source	Enr.↑	pLL↑	Edit	DCA↓
Held-out leaves	–	–	–	–
PROPHET	–	–	–	–
ESM-only	–	–	–	–
Random mut.	–	–	–	–

For all MOG-DFM methods we generate 500 peptides of length 10 per condition, sampling weight vectors ω uniformly from Δ^1 to cover the Pareto front.

4.1 Experiment 1: Variant Quality

We first validate that the phylogenetic energy function produces biologically realistic escape variants. We sample $M = 500$ variants from p_{evo} via Gibbs sampling at the calibrated T_{evo} and compare against three variant generation strategies: (i) uniform random mutations, (ii) ESM-only iterative masking, and (iii) observed leaf sequences from held-out phylogenies.

For each method, we evaluate: (a) *resistance site enrichment*, the fraction of mutated positions overlapping with documented HIV-1 protease resistance positions from the Stanford HIV Drug Resistance Database; (b) *ESM pseudo-log-likelihood*, as a proxy for structural viability; (c) *edit distance distribution*, comparing against held-out leaf sequences; and (d) *DCA energy*, measuring consistency with the learned coevolutionary landscape.

We expect PROPHET variants to be significantly enriched at known resistance positions compared to random and ESM-only sampling, while maintaining structural plausibility comparable to observed leaves.

Table 2. Escape-robust peptide design on HIV-1 protease.

Method	WT ↑	Mean ↑	Min ↑	Ret. ↑
RFdiffusion	–	–	–	–
MOG-DFM (WT)	–	–	–	–
MOG-DFM (rand.)	–	–	–	–
MOG-DFM (leaves)	–	–	–	–
MOG-DFM (ESM)	–	–	–	–
PepTune	–	–	–	–
PROPHET	–	–	–	–

Table 3. Pareto front quality.

Method	HV ↑	Pareto ct. ↑
RFdiffusion	–	–
MOG-DFM (WT)	–	–
MOG-DFM (leaves)	–	–
MOG-DFM (ESM)	–	–
PepTune	–	–
PROPHET	–	–

4.2 Experiment 2: Escape-Robust Design

We evaluate whether phylogenetically informed robustness guidance yields peptides that maintain binding across escape variants. Using a held-out set of known HIV-1 protease resistance mutants not seen during phylogenetic estimation, we score each designed peptide on four metrics: (i) *wildtype binding* (WT); (ii) *mean escape binding* across all held-out variants; (iii) *minimum escape binding* against the hardest variant; and (iv) *binding retention* (Ret.), the fraction of variants above threshold τ_{bind}.

We hypothesize that PROPHET will achieve significantly higher mean and minimum escape binding than all baselines, with a modest reduction in wildtype binding reflecting the trade-off between single-target optimization and distributional robustness. We expect RFdiffusion and MOG-DFM (WT) to score highest on wildtype binding but drop sharply on escape variants.

4.3 Experiment 3: Pareto Front Analysis

To characterize the robustness–affinity trade-off, we visualize the Pareto front of wildtype binding (s_1) versus mean escape binding for each method. We additionally report the hypervolume indicator (HV), which measures the area dominated by the front relative to a reference point.

4.4 Experiment 4: Ablations

We isolate the contribution of each PROPHET component by systematically removing or replacing individual design choices. All ablations generate 500 peptides under identical sampling conditions and are evaluated on the held-out escape variant set.

Table 4. Ablations on HIV-1 protease.

Setting	WT ↑	Mean ↑	Min ↑	Ret. ↑
Full PROPHET	–	–	–	–
− DCA (J_{ij}=0)	–	–	–	–
− λ_i wt.	–	–	–	–
− ESM filter	–	–	–	–
− Gibbs (leaves)	–	–	–	–
CVaR η=1.0	–	–	–	–
CVaR η=0.5	–	–	–	–
CVaR η=0.1	–	–	–	–

Table 5. Sensitivity to T_{evo}.

T_{evo}	Edit	Mean ↑	Min ↑	Ret. ↑
0.5	–	–	–	–
1.0	–	–	–	–
2.0	–	–	–	–
5.0	–	–	–	–

Phylogenetic energy components. We ablate three components of E_{evo}: (a) removing DCA couplings by setting $J_{ij} = 0$, reducing the model to independent site sampling; (b) removing mutation-rate weighting by setting $\lambda_i = 1$, treating all sites equally; and (c) removing the ESM filter by setting $\delta = \infty$, accepting all Gibbs samples regardless of structural viability. We additionally compare against using observed leaf sequences directly (bypassing the energy model entirely).

CVaR stringency. We vary $\eta \in \{0.1, 0.5, 1.0\}$ to evaluate the trade-off between mean-case and worst-case robustness.

4.5 Experiment 5: Sensitivity Analysis

Evolutionary temperature. We vary $T_{\text{evo}} \in \{0.5, 1.0, 2.0, 5.0\}$ to evaluate sensitivity to the breadth of the variant distribution.

Number of sampled variants. We vary $M \in \{50, 100, 250, 500, 1000\}$ to assess how many Gibbs samples are needed for the robustness objective to stabilize.

Number of phylogenies. We subsample $J \in \{25, 50, 100, 200\}$ to evaluate how many trees are needed for stable energy function estimation.

5 Discussion

We have presented **PROPHET**, the first framework to integrate phylogenetic evolutionary modeling directly into the generative objective of discrete flow matching for peptide design. The central methodological contribution is the phylogenetic energy function (Eq. 4), which translates ob-

Table 6. Sensitivity to M. Var. is variance of s_{robust} across 10 runs.

M	Var.↓	Mean↑	Min↑	Ret.↑
50	–	–	–	–
100	–	–	–	–
250	–	–	–	–
500	–	–	–	–
1000	–	–	–	–

Table 7. Sensitivity to J.

J	Mean↑	Min↑	Ret.↑
25	–	–	–
50	–	–	–
100	–	–	–
200	–	–	–

served evolutionary dynamics into a structured distribution over future viral variants usable as a guidance signal for any pre-trained discrete generative model. This represents a qualitatively different design paradigm from existing approaches: rather than optimizing against a fixed target and hoping the result generalizes, PROPHET optimizes against the distribution of targets evolution is likely to produce, as estimated from the virus's own phylogenetic history. The CVaR robustness objective further allows practitioners to interpolate smoothly between average-case and worst-case evolutionary robustness within a single unified framework. Practically, PROPHET requires no new experimental data, no learned tree generator, and no adversarial training. The energy function is estimated from standard FastTree (Price et al., 2009) phylogenies and DCA, both routinely computed in viral genomics pipelines, and the robustness objective integrates into MOG-DFM (Chen et al., 2025c) as a drop-in guidance signal with no architectural modification, making PROPHET immediately deployable for any viral target with sufficient phylogenetic coverage.

The framework has limitations worth noting: the energy function is estimated from a finite tree ensemble over a single protein family, robustness holds insofar as future evolution remains within the support of the estimated distribution, and the ESM plausibility filter is less characterized for heavily glycosylated surface antigens, which occur frequently, but can be ameliorated with post-translational modification-aware langugage models (Peng et al., 2025). Future work will extend PROPHET to multi-protein targets where escape involves coordinated mutations across interacting viral genes and validate designed peptides experimentally against clinically observed resistance mutants. Nonetheless, PROPHET establishes that the phylogenetic record of a pathogen contains sufficient information to guide generative design of therapeutics that anticipate evolutionary escape, opening a path toward antiviral peptides that remain effective across the evolutionary lifetime of a virus rather than a single snapshot of it.

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Appendix

A Algorithms

Algorithm 1 Phylogenetic Energy Estimation and Variant Sampling

Input: wildtype sequence $x_{\mathrm{WT}} \in \mathcal{T}^N$
    FastTree phylogenies $\{T_j\}_{j=1}^{J}$ over homologous sequences
    multiple sequence alignment $\mathcal{A}$
    ESM-2 model p_{ESM}
Hyperparameters: evolutionary temperature T_{evo}, ESM filter threshold δ, number of samples M, burn-in steps B
▷ *Estimate site-specific evolutionary parameters*
for each position $i = 1$ to N **do**
    Estimate mutation rate λ_i from $\{T_j\}$ ▷ *substitutions / total branch length*
    Estimate substitution matrix Q_i from $\{T_j\}$ ▷ *empirical transition frequencies*
end for
▷ *Estimate pairwise couplings via DCA*
Fit Potts model to $\mathcal{A}$ via pseudolikelihood maximization
Extract fields $\{h_i\}_{i=1}^{N}$ and couplings $\{J_{ij}\}_{i<j}$
▷ *Construct phylogenetic energy function*
$E_{\mathrm{evo}}(x) \leftarrow \sum_i \lambda_i \, h_i(x_i) + \sum_{i<j} J_{ij}(x_i, x_j)$
▷ *Gibbs sampling with ESM filter*
Initialize $x \leftarrow x_{\mathrm{WT}}$
$\mathcal{V} \leftarrow \emptyset$
for $t = 1$ to $B + M$ **do**
    for $i = 1$ to N **do** ▷ *full Gibbs sweep*
        Sample $x_i \sim p_{\mathrm{evo}}(x_i \mid x_{\setminus i})$ ▷ *Eq. 6*
    end for
    if $t > B$ **then** ▷ *past burn-in*
        Compute $\mathrm{pLL}(x) = \sum_i \log p_{\mathrm{ESM}}(x_i \mid x_{\setminus i})$
        if $\mathrm{pLL}(x) \geq \mathrm{pLL}(x_{\mathrm{WT}}) - \delta$ **then**
            $\mathcal{V} \leftarrow \mathcal{V} \cup \{x\}$
        end if
    end if
end for
return variant set $\mathcal{V}$, energy function E_{evo}

Algorithm 2 **Escape-Robust Peptide Sampling via MOG-DFM**

Input: pre-trained discrete flow model u_{θ_0}
    wildtype target x_{WT}
    sampled variant set $\mathcal{V} = \{x^{(m)}\}_{m=1}^{M}$ from Algorithm 1
    binding affinity predictor $\mathrm{Aff}(\cdot, \cdot)$
    desired peptide length L
Hyperparameters: CVaR level η, guidance strength β, weight vector $\omega \in \Delta^1$, hypercone angle Φ, step size Δt
▷ *Define multi-objective guidance*
$s_1(y) \leftarrow \mathrm{Aff}(y,\, x_{\mathrm{WT}})$ ▷ *wildtype binding*
$s_2(y) \leftarrow \frac{1}{\lfloor \eta M \rfloor} \sum_{m=1}^{\lfloor \eta M \rfloor} \mathrm{Aff}(y,\, x^{(\sigma(m))})$ ▷ *CVaR robustness*
▷ *MOG-DFM reverse sampling*
Initialize peptide $x_1 \sim p_0$ ▷ *noised initial state*
for $t = 1, 1 - \Delta t, \ldots, \Delta t$ **do**
    for each position $i \in \{1, \ldots, L\}$ **do**
        for each candidate token $y^i \in \mathcal{T} \setminus \{x_t^i\}$ **do**
            Construct candidate $x_{\mathrm{new}} \leftarrow x_t$ with $x_{\mathrm{new}}^i = y^i$
            Evaluate $s_1(x_{\mathrm{new}})$ and $s_2(x_{\mathrm{new}})$
            Compute $\Delta S(y^i, x_t, \omega)$ from rank scores and directional alignment
        end for
        Apply hypercone filter with angle Φ_t around ω
        Compute guided rate $u_{\mathrm{guided},t}^{i}(y^i, x_t \mid \omega)$ ▷ *Eq. 2*
    end for
    Sample transition $x_{t-\Delta t}$ from guided rates
end for
return peptide $y^* \leftarrow x_0$ ▷ *escape-robust binder*